THE ISLAMIA UNIVERSITY OF BAHAWALPUR

Department of Information Technology

PROJECT PROPOSAL

Smart Warehouse Distribution & Inventory Management System

A Modern SaaS-Based Warehouse, Inventory, Order and Distribution Management Solution

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1. Introduction

The **Smart Warehouse Distribution & Inventory Management System** is a modern web-based SaaS application designed to digitize and streamline the operations of a large-scale wholesale warehouse. Such warehouses typically manage products from multiple companies and categories, receive stock periodically, collect orders from retail shops through sales representatives, and distribute the required products on scheduled deliveries.

The proposed system will provide a centralized platform for managing companies, categories, products, product images, inventory, retail shops, salesmen, orders, deliveries, users, notifications and reports. It will provide a modern administrative dashboard for monitoring stock levels, order activities, sales trends and overall warehouse performance in a clear and visual manner.

2. Problem Statement

Traditional warehouse and distribution operations often depend on manual registers, spreadsheets, phone calls and disconnected records. These approaches can make it difficult to maintain accurate stock information, track shop orders, monitor stock movement and generate timely reports. Manual processes can also lead to duplicate entries, calculation errors, delayed order processing and limited visibility for management.

Therefore, there is a need for a centralized, secure and scalable software solution that can manage warehouse inventory and distribution activities efficiently while providing real-time information to authorized users.

3. Project Objectives

- Develop a centralized system for managing warehouse products, companies, categories and inventory.
- Maintain accurate records of incoming, outgoing, available, damaged and expired stock.
- Provide a dedicated shop management system where each retail shop has its own account and order history.
- Allow salesmen to create and submit shop orders efficiently from the field.
- Provide role-based access for administrators, managers, salesmen and viewers.
- Develop a modern dashboard with charts, KPIs and visual analytics for warehouse operations.
- Provide real-time notifications and updates for important order and stock activities.
- Generate professional PDF and Excel reports for inventory, orders and sales.
- Support product images so users can visually identify products during product management and order creation.
- Provide light and dark themes and a responsive user interface.
- Build the system using scalable enterprise software architecture and industry-standard development practices.

4. Proposed System / Scope

The proposed system will cover the complete workflow from product and stock management to shop order collection and distribution. Administrators will be able to add companies, categories and products, including product images and relevant details. Warehouse staff will maintain inventory and record incoming and outgoing stock. Salesmen will access assigned shops and create orders based on customer requirements.

Administrators and managers will review, approve, process and monitor orders through the system.

- **Company Management:** Add, edit, view and manage product companies.
- **Category Management:** Manage product categories such as beverages, snacks, biscuits and grocery items.
- **Product Management:** Manage product details, pricing, stock, expiry information and images.
- **Inventory Management:** Track incoming, available, outgoing, damaged and expired stock.
- **Shop Management:** Maintain individual shop profiles and order histories.
- **Salesman Management:** Manage sales representatives and their assigned shops.
- **Order Management:** Create, review, approve, process, dispatch and deliver orders.
- **Reports:** Generate inventory, sales, shop and order reports in PDF/Excel formats.
- **Notifications:** Provide alerts for new orders, low stock and other important activities.
- **Dashboard:** Present real-time KPIs and analytics through modern charts.
- **User & Role Management:** Control access according to user roles and permissions.

5. Proposed Methodology

The project will follow an **iterative and modular software development methodology**. The system will be divided into independent modules and developed incrementally. Each module will be analyzed, designed, implemented, tested and integrated before moving to the next major module. This approach will make the project easier to manage, test and improve.

- **Requirement Analysis:** Identify business requirements, users, workflows and system constraints.
- **System Design:** Define architecture, modules, database relationships and user roles.
- **Database Design:** Create normalized SQL Server tables, relationships, keys and indexes.
- **Backend Development:** Develop secure REST APIs using ASP.NET Core 8 Web API.
- **Frontend Development:** Build a responsive Angular 21 standalone application using Tailwind CSS.
- **Integration:** Connect Angular frontend with backend APIs and real-time SignalR services.
- **Testing:** Perform functional, integration, validation, authorization and usability testing.
- **Deployment:** Prepare the application for production deployment and future scalability.

6. Tools & Technologies

Area	Technology
Frontend	Angular 21 Standalone, TypeScript, Tailwind CSS, Angular Material
Charts & Analytics	ApexCharts / ng-apexcharts
Backend	ASP.NET Core 8 Web API
Database	Microsoft SQL Server
ORM	Entity Framework Core
Architecture	Clean Architecture, Repository Pattern, Service Pattern, Unit of Work
Data Transfer	DTOs with AutoMapper

Area	Technology
Validation	FluentValidation
Authentication	JWT Authentication & Role-Based Authorization
Real-Time	ASP.NET Core SignalR
Reporting	PDF and Excel generation/export
Development Tools	Visual Studio / VS Code, Git & GitHub

7. System Architecture

The application will follow a layered enterprise architecture. The Angular 21 frontend will communicate with the ASP.NET Core 8 Web API. The backend will use DTOs, services, repositories and Unit of Work to separate business logic from data access. Entity Framework Core will communicate with SQL Server. SignalR will provide real-time updates and notifications.

- **Presentation Layer:** Angular 21 Standalone application.
- **API Layer:** ASP.NET Core Web API controllers and middleware.
- **Application Layer:** DTOs, services, validation and business logic.
- **Domain Layer:** Entities and core business rules.
- **Infrastructure Layer:** EF Core, repositories, Unit of Work and database access.
- **Database:** SQL Server with relational constraints and optimized indexes.

8. Key Functional Modules

- Authentication & Authorization
- Dashboard & Analytics
- Company Management
- Category Management
- Product Management
- Product Image Management
- Warehouse & Inventory Management
- Shop Management
- Salesman Management
- Order Management
- Delivery & Dispatch Management
- Reports & Export
- Notifications
- User & Role Management
- Settings & Profile
- Audit Logs

9. Expected Outcomes

- A centralized digital platform for warehouse and distribution operations.
- Improved accuracy of inventory and order records.
- Faster shop order collection and processing.
- Better visibility of stock availability and stock movement.
- Reduced dependence on manual paperwork and spreadsheets.
- Modern dashboards for data-driven management decisions.
- Real-time notifications for important warehouse activities.
- Professional reports for management and operational analysis.
- Secure role-based access to sensitive business information.
- A scalable foundation that can support multiple warehouses and tenants in the future.

10. Future Enhancements

- Mobile application for salesmen and delivery staff.
- Barcode and QR code scanning.
- Online payment and digital invoice integration.
- Advanced demand forecasting and inventory prediction using AI.
- Route optimization for deliveries.
- Integration with SMS and WhatsApp notification services.
- Advanced multi-tenant subscription and billing management.
- Business intelligence and advanced predictive analytics.

11. Conclusion

The **Smart Warehouse Distribution & Inventory Management System** will provide a complete digital solution for managing the major activities of a wholesale warehouse and its retail distribution network. By combining modern web technologies, structured software architecture, real-time communication, visual analytics and secure role-based access, the proposed system will improve operational efficiency, inventory visibility and order management. The project will also provide a strong practical implementation of modern full-stack software development concepts and can be extended into a larger commercial SaaS platform in the future.

12. Proposed Development Plan

Phase	Major Activity
01	Requirement Analysis & System Planning
02	Database & ERD Design
03	Backend Architecture & Project Setup
04	Authentication & Authorization
05	Company, Category & Product Management
06	Product Images & Inventory Management

Phase	Major Activity
07	Shop & Salesman Management
08	Order & Delivery Management
09	Dashboard & Analytics
10	Reports, Notifications & Real-Time Features
11	Testing & Security
12	Deployment & Final Documentation

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